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Modeling the spread of the Corona virus

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This study develops an agent-based SEIR simulation of COVID-19 transmission within the PyCX framework, calibrated with literature-informed epidemiological parameters, and integrates a genetic algorithm (PyGAD) to identify near-optimal intervention strategies. Seven scenarios are evaluated over a population of 500 agents on an 80-day horizon: a baseline, four single-mechanism interventions (strict lockdown, high mask compliance, social distancing modelled as a reduced effective contact radius, and vaccination via pre-existing immunity), a combined multi-layered intervention, and a GA-optimised parameter set. Each scenario is averaged over 30 independent simulation seeds, and the GA is executed thirty times to quantify robustness. The combined intervention reduces cumulative infections from a baseline of 494.9 ± 0.3 to 3.3 ± 3.2 , and the GA-identified optimum reduces them further to 0.2 ± 0.3 . A per-gene robustness analysis reveals an asymmetric convergence pattern in which the contact-related parameters β and r converge tightly while the mobility and immunity parameters admit a wider range of near-equivalent solutions, indicating partial substitutability between lockdown intensity and vaccination coverage along an optimal-policy ridge.

CCS Concepts: • **Computing methodologies** → **Agent / discrete models**; **Network science**; **Uncertainty quantification**.

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1 Introduction

Coronavirus disease 2019 (COVID-19) is an infectious respiratory illness caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). SARS-CoV-2 was first identified in December 2019 in Wuhan, China, after reports of a pneumonia cluster of unknown origin were submitted to the World Health Organization (WHO) [Chauhan 2020; Harapan et al. 2020].

Coronaviruses are a group of RNA viruses that infect mammals and birds, with several species responsible for respiratory illnesses in humans. Previous outbreaks caused by related coronaviruses include Severe Acute Respiratory Syndrome (SARS) in 2003 and Middle East Respiratory Syndrome (MERS) in 2012 [Hamed 2020].

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Computational modeling is a valuable tool for simulating the spread of infectious diseases within populations and for evaluating potential mitigation strategies. This study develops a complex system simulation to model the spread of COVID-19 using realistic epidemiological parameters, including infection rate, recovery rate, and transmission probability. Systematic variation of these parameters enables investigation of how different interventions or behavioral changes may influence epidemic progression. The objective is to analyze the effectiveness of such strategies in reducing viral transmission and to provide insights into approaches for mitigating future outbreaks.

2 Related work

Recent research on COVID-19 modeling encompasses a broad spectrum of methodologies, ranging from classical compartmental models to advanced simulation-based techniques. While traditional models such as SI, SIR, and SEIR are widely used for their simplicity, they often do not adequately represent the complex and heterogeneous dynamics of real-world epidemic spread [Stanovov et al. 2022]. To overcome these limitations, modified and hybrid models have been developed, including the application of evolutionary algorithms for parameter calibration, which substantially enhance model fitting and the identification of underlying transmission mechanisms [Stanovov et al. 2022].

Agent-based models (ABMs) and multi-agent simulations have become prominent tools for modeling COVID-19 transmission at the individual level. These methods facilitate the representation of interactions among individuals and support the evaluation of intervention strategies, including social distancing, testing, and reopening policies [Rodriguez et al. 2022]. Additionally, multi-agent models can account for heterogeneous populations and nonlinear interactions, making them well-suited for investigating complex epidemic dynamics. However, these models also present challenges related to validation and computational efficiency [Castro et al. 2022; Jin et al. 2022].

Beyond agent-based approaches, cellular automata (CA) models have been extensively utilized to capture the spatial dimensions of epidemic spread. These models simulate local interactions among individuals or regions and can incorporate spatial dependencies, including population density and mobility patterns [Podolski and Nguyen 2021]. Previous research has shown that integrating network structures, such as small-world effects, can significantly accelerate disease transmission and enhance the realism of epidemic simulations [GAGLIARDI and ALVES 2010]. Recent studies have combined CA with agent-based modeling to develop hybrid frameworks that integrate spatial dynamics with individual behavior, thereby increasing the accuracy of spatio-temporal predictions [Wang et al. 2024].

Optimization techniques, particularly evolutionary algorithms, have been employed not only for model calibration but also for intervention planning, including vaccine distribution and resource allocation. These methods contribute to more efficient epidemic control strategies and facilitate faster attainment of herd immunity [Zheng et al. 2023]. Collectively, the literature indicates that integrating realistic data, agent-based or spatial modeling, and optimization techniques establishes a comprehensive framework for analyzing and mitigating virus transmission. This integrated approach underpins the methodology adopted in the present project.

3 Methods and Materials

An Agent-Based Model (ABM) was developed using the PyCX framework to simulate the transmission dynamics of COVID-19. In contrast to traditional compartmental models that employ ordinary differential equations to represent population averages, the ABM approach represents each individual as an autonomous entity with defined states and behaviors. This methodology enables the emergence of complex patterns resulting from local interactions rather than relying on global assumptions.

3.1 Model Architecture: The SEIR Framework

- **Susceptible (S):** Healthy individuals who have not yet been infected but are vulnerable to the virus upon contact with a carrier [Chauhan 2020].
- **Exposed (E):** Individuals who have been infected but are in the "latent period." This state accounts for the documented incubation period, which typically lasts between 2 to 14 days, during which the individual is not yet showing symptoms or necessarily infectious [Hamed 2020; Harapan et al. 2020].
- **Infectious (I):** Individuals capable of transmitting the virus. Based on clinical evidence that SARS-CoV-2 is primarily transmitted through respiratory droplets and close human-to-human contact [Hamed 2020; Harapan et al. 2020], this state is modeled using a specific spatial radius to represent the physical proximity required for transmission [Yasmin and Ansari 2025].
- **Recovered (R):** Individuals who have cleared the infection. This state assumes the development of immunity, preventing the individual from re-entering the transmission chain [Chauhan 2020; Yasmin and Ansari 2025].

3.2 Mathematical Parameters and Formulas

The transition dynamics of our agent-based model are driven by specific epidemiological parameters sourced from clinical literature. These parameters define the transition probabilities between the states of Susceptible (S), Exposed (E), Infectious (I), and Recovered (R).

3.2.1 Parameter Values from Literature. The following values serve as the foundation for the simulation's realism:

- **Basic Reproduction Number (R_0):** The model uses an R_0 range of 2.2 to 2.68, representing the initial outbreak intensity [Harapan et al. 2020].

- **Incubation Period (τ_{inc}):** To model the Exposed (E) state, we use a mean incubation period of 5.2 days [Harapan et al. 2020], within a clinical range of 2 to 14 days [Chauhan 2020; Hamed 2020].
- **Recovery Period (τ_{rec}):** The transition from Infectious (I) to Recovered (R) is based on an average recovery time of 10 to 14 days for typical cases [Yasmin and Ansari 2025].
- **Transmission Radius (r):** Based on evidence that the virus spreads via respiratory droplets [Hamed 2020], we implement a spatial radius where infection occurs if a Susceptible agent is within distance r of an Infectious agent.

3.2.2 Transition Rates and Probabilities. In the simulation code, time-based durations are converted into daily transition probabilities. The rate of change for each compartment is modeled as follows:

$$\sigma = \frac{1}{\tau_{inc}} \approx \frac{1}{5.2} \approx 0.192 \quad (1)$$

Where σ is the probability per time step of an agent moving from the **Exposed** to the **Infectious** state.

$$\gamma = \frac{1}{\tau_{rec}} \approx \frac{1}{10} = 0.1 \quad (2)$$

Where γ is the probability per time step of an agent moving from the **Infectious** to the **Recovered** state.

The **Transmission Probability (β)** is treated as our primary **tuning variable**. While the baseline is derived from R_0 , we vary β systematically to investigate the impact of non-pharmaceutical interventions (NPIs) like mask-wearing or social distancing.

3.2.3 Summary of Model Calibration. [H]

Parameter	Symbol	Value	Primary Source
Basic Reproduction Number	R_0	2.2 – 2.68	[Harapan et al. 2020]
Mean Incubation Time	τ_{inc}	5.2 days	[Hamed 2020; Harapan et al. 2020]
Mean Infectious Period	τ_{rec}	10 days	[Yasmin and Ansari 2025]
Transmission Probability	β	Tuned Variable	[Chauhan 2020]

Table 1. Epidemiological parameters used for model initialization and testing.

3.3 Spatial Environment and Movement

The simulation environment is a two-dimensional continuous space in which agents follow a random walk. At each time step, an agent's position (x, y) is updated by a small random displacement. This movement serves as a simplified model of human mobility and influences the frequency of contact events, thereby affecting the likelihood of transmission.

3.4 Tools and Software

The simulation and optimization components of this project were developed using a high-level technical stack centered around the Python programming language.

- **Programming Language: Python 3.x** was chosen because it supports scientific computing well and is flexible enough to handle complex agent-based logic.
- **Simulation Framework: PyCX** was used as the main simulation framework. It offers a real-time environment for modeling complex systems and manages the update loops and agent states.
- **Numerical Computation: NumPy** was used for efficient array operations and random processes, like generating random walk steps and checking distances for infections.
- **Visualization: Matplotlib** was the main tool for real-time visualization. It animated agent movements and plotted SEIR curves showing population changes over time.
- **Optimization Algorithm: A Genetic Algorithm (GA)** was used to optimize parameters. This method helped the model find the best balance between transmission probability (β) and agent mobility for simulating policy decisions.

4 Experimental Procedure

4.1 Simulation Scenarios and Parameter Sets

To evaluate the effectiveness of different interventions, we defined seven distinct scenarios. All scenarios use a population size of $N = 500$ and a transmission radius of $r = 0.05$.

- (1) **Scenario 1: Baseline (Pre-Intervention Conditions)** Scenario 1 simulates early pandemic conditions before strong public health interventions were introduced. Mobility remains high ($\epsilon = 0.03$), representing normal population movement prior to major restrictions. This assumption is supported by empirical evidence showing that human mobility remained close to or above baseline levels before lockdown measures were implemented [Hadjidemetriou et al. 2020].

The transmission probability is kept at a relatively high level ($\beta = 0.5$), reflecting limited or inconsistent mask use in the community during the early phase of the COVID-19 pandemic, when many public health authorities had not yet recommended universal masking [MacIntyre and Chughtai 2020].

- (2) **Scenario 2: Strict Lockdown with Reduced Mobility** Scenario 2 models a strict lockdown by reducing the mobility parameter. The parameter is adjusted from $\epsilon = 0.03$ to $\epsilon = 0.01$. This change represents a substantial reduction in population movement, consistent with empirical observations during COVID-19 containment policies.

Studies utilizing transport and mobility datasets have reported reductions of approximately 60 to 80 percent in travel activity following lockdown interventions [Hadjidemetriou et al. 2020]. Mobile-phone tracking analyses in Italy observed mobility decreases ranging from 44 percent to 63 percent after the implementation of stricter restrictions. These reductions were associated with earlier epidemic control and delayed infection peaks [Vinceti et al. 2020].

Additionally, reductions in social distancing behavior have been shown to correlate with lower infection growth rates and reduced transmission intensity across regions [Shi et al. 2025]. Therefore, the reduced value of ϵ is considered a literature-informed threshold that represents strong mobility suppression rather than a directly measured empirical constant.

- (3) **Scenario 3: High Mask Compliance with Reduced Transmission Probability (β)** Scenario 3 simulates high mask compliance by reducing the transmission probability from $\beta = 0.5$ to $\beta = 0.1$, while maintaining high mobility ($\epsilon = 0.03$). This adjustment represents a substantial reduction in effective transmission, consistent with empirical evidence regarding mask use during the COVID-19 pandemic.

Systematic reviews and meta-analyses have demonstrated that face masks are associated with a substantial reduction in infection risk across healthcare and community settings [Chu et al. 2020; Li et al. 2021]. Specifically, mask use has been reported to significantly reduce the risk of SARS-CoV-2 infection, with stronger protective effects observed for respirators such as N95 masks compared to standard surgical masks [Chu et al. 2020].

Additional review evidence indicates that community mask use can be beneficial, particularly in contexts involving pre-symptomatic transmission, and that masks used for source control may help protect close contacts [MacIntyre and Chughtai 2020]. Consequently, the reduced value of β is interpreted as a literature-informed threshold representing strong masking effectiveness rather than a directly measured empirical constant.

- (4) **Scenario 4: Social Distancing with Reduced Effective Contact Radius (r)** Scenario 4 simulates physical social distancing by reducing the effective transmission radius from $r = 0.05$ to $r = 0.02$, while maintaining the per-contact transmission probability ($\beta = 0.5$) and mobility ($\epsilon = 0.03$) at baseline levels. The parameter r specifies the spatial zone within which a susceptible agent may acquire infection from an infectious neighbor, reflecting the physical proximity necessary for respiratory droplet exposure. Reducing r thus represents the behavioral effect of individuals maintaining greater interpersonal distance, narrowing the effective window for transmission without modifying either per-contact biological risk (as in masking, Scenario 3) or aggregate mobility (as in lockdown, Scenario 2).

This approach aligns with empirical and theoretical findings on close-contact droplet transmission. [Qian and Jiang 2020] emphasizes that maintaining approximately 1.5 m between individuals is sufficient to prevent most respiratory infections, as droplets from coughing, sneezing, or speaking travel a limited distance that can largely be avoided through spacing. [Sun and Zhai 2020] extends the classical Wells–Riley infection model by incorporating an explicit distance index and derives safe conversational distances of 1.6 to 3.0 m. Their results indicate that roughly doubling the effective interpersonal distance reduces infection probability by approximately 20 to 40 percent during short exposures. The radius-reduction method is also methodologically consistent with [Maheshwari and Albert 2020], where

social distancing is implemented in a network-SIR framework by removing non-essential interaction edges between individuals, reducing the effective reproduction number R_t from approximately 6 to below 1. Although that model modifies graph edges rather than a spatial radius, the underlying abstraction, namely reducing the effective contact structure rather than per-contact risk, remains consistent.

Population-level empirical evidence further substantiates this intervention. [de Souza Melo et al. 2021] reports that a 1 percent increase in a mobile-phone-based isolation index in São Paulo corresponded to a 6.91 percent reduction in daily new cases and a 6.90 percent reduction in deaths, with 30-day cumulative reductions reaching 22.72 and 35.39 percent, respectively. Similarly, [Thu et al. 2020] finds that stricter, earlier social distancing measures across 10 severely affected countries led to more rapid declines in daily case counts within 1 to 4 weeks of implementation. Cross-country econometric analyses further demonstrate that containment and social-distancing interventions significantly reduced the growth rate of COVID-19 infections across multiple countries, reinforcing the interpretation that limiting effective interpersonal contact distance reduces transmission opportunities at the population level [Prakash et al. 2022]. Collectively, these findings justify treating the reduction of r as a literature-informed approximation of strong interpersonal distancing compliance, rather than as a directly measured empirical constant.

- (5) **Scenario 5: Vaccination Campaign with Initial Population Immunity** Scenario 5 simulates a vaccination campaign by assigning 40% of the population to the Recovered (R) compartment at the start of the simulation to represent individuals with effective immunity from vaccination and/or prior infection. This adjustment reflects population-level immunity observed during later phases of the COVID-19 pandemic, when substantial fractions of individuals had acquired protection through vaccination programs or previous exposure.

Clinical trials reported initial vaccine efficacy against symptomatic infection ranging from approximately 66% to 95%, while prior infection has been estimated to provide around 87% protection against reinfection over several months [Pooley et al. 2023]. Additional evidence suggests that hybrid immunity, combining vaccination and prior infection, provides more durable protection than either source alone and contributes significantly to reductions in transmission and severe disease risk [Moore et al. 2025]. Booster vaccination has also been shown to further increase protection against severe outcomes, with effectiveness exceeding 70% in pooled analyses [Song et al. 2024].

Therefore, the assumption that 40% of the population begins in the Recovered state is interpreted as a literature-informed approximation of partial pre-existing immunity rather than a strict herd immunity threshold.

- (6) **Scenario 6: Combined Real-World Intervention** Scenario 6 models a combined intervention strategy that integrates reduced transmission probability, decreased agent mobility, and

partial population immunity. This approach reflects the layered public health responses implemented during the COVID-19 pandemic. The transmission probability is set to $\beta = 0.1$, aligning with empirical evidence that high mask compliance significantly lowers the risk of SARS-CoV-2 infection in both community and healthcare settings [Chu et al. 2020; Li et al. 2021; MacIntyre and Chughtai 2020]. Agent mobility is reduced to $\varepsilon = 0.01$, based on studies indicating mobility reductions of 60 to 80 percent following government-imposed lockdowns [Hadjidemetriou et al. 2020; Shi et al. 2025; Vinceti et al. 2020]. Furthermore, 40 percent of the population is initialized in the Recovered compartment to represent individuals with pre-existing immunity from vaccination or prior infection, consistent with population-level immunity estimates from later pandemic phases [Moore et al. 2025; Pooley et al. 2023; Song et al. 2024]. This scenario provides a literature-informed approximation of real-world multi-layered interventions and serves as a direct reference for evaluating the genetic algorithm optimization results presented in Scenario 7.

- (7) **Scenario 7: Genetic Algorithm Optimization** Scenario 7 utilizes a Genetic Algorithm (GA), implemented with the PyGAD library, to identify the optimal combination of intervention parameters that minimizes total cumulative infections in the simulation. This methodology aligns with previous research demonstrating the effectiveness of evolutionary algorithms for calibrating epidemiological models and optimizing intervention strategies [Stanovov et al. 2022; Zheng et al. 2023]. The GA represents three parameters as genes: transmission probability (β), agent mobility (ε), and the initial fraction of immune individuals, which reflects vaccination coverage. Parameter search bounds are restricted to epidemiologically plausible ranges based on the literature: $\beta \in [0.05, 0.5]$, $\varepsilon \in [0.005, 0.03]$, and $\text{immune_fraction} \in [0.0, 0.6]$. The fitness function is defined as the minimization of the total cumulative number of agents passing through the Infectious compartment by the end of the simulation, as this metric directly reflects the overall disease burden [Stanovov et al. 2022]. Each individual in the GA population represents a unique parameter combination, evaluated by running a complete agent-based SEIR simulation and returning the resulting infection count as the fitness score. The GA iterates across multiple generations using selection, crossover, and mutation to converge on the parameter combination that produces the lowest total infections. The GA-optimized parameters are then compared with the literature-informed values established in Scenario 6 to quantify the difference between empirically observed real-world intervention strategies and the near-optimal strategy identified through evolutionary optimization [Stanovov et al. 2022; Zheng et al. 2023].

5 Results

The agent-based SEIR model was evaluated across seven distinct scenarios, described in Section 4. For Scenarios 1 through 6, the spatial distribution of the agent population is visualised at $t = 1$ (initial configuration) and $t = 100$ (end of simulation). The parameter settings for each fixed-parameter scenario are summarised in Table 2, and the corresponding aggregate metrics are averaged over

30 independent simulation seeds. Scenario 7 reports the outcome of the genetic algorithm optimisation across 30 independent GA runs.

5.1 Scenario 1: Baseline

Figure 1 presents the initial and final spatial configuration of the agent population under baseline conditions ($\beta = 0.5$, $\varepsilon = 0.03$, $r = 0.05$, with no initial immunity). At $t = 1$, the population is uniformly susceptible (cyan) with five infectious seeds (red). By $t = 100$ the population has almost completely transitioned to the Recovered state (green), with only a negligible number of susceptibles remaining. The mean total cumulative infections reach 494.9 ± 0.3 out of 500 agents ($\approx 99\%$), and the infectious peak occurs at $t \approx 21.5$ days, reaching 177.2 ± 17.7 simultaneously infectious individuals. The very low standard deviation on the total count confirms that, in the absence of any intervention, the epidemic propagates deterministically through almost the entire population, matching the basic reproduction number range ($R_0 = 2.2\text{--}2.68$) reported in [Harapan et al. 2020].

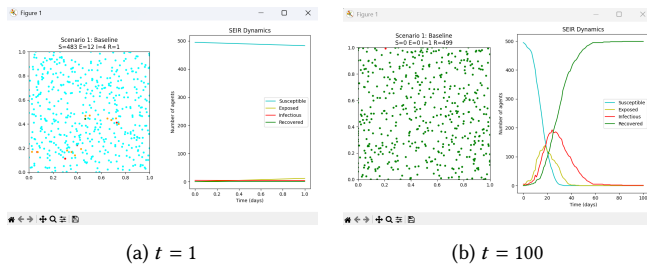


Fig. 1. Scenario 1 (Baseline, $\beta = 0.5$, $\varepsilon = 0.03$, $r = 0.05$, $\phi = 0$): spatial state at $t = 1$ (left) and $t = 100$ (right). Cyan: Susceptible, orange: Exposed, red: Infectious, green: Recovered.

5.2 Scenario 2: Strict Lockdown

A reduction in mobility from $\varepsilon = 0.03$ to $\varepsilon = 0.01$ (Figure 2) results in a slower diffusion of infectious agents through the spatial domain. At $t = 100$, a substantially larger fraction of agents remains susceptible (cyan) compared to the baseline. However, a substantial fraction of recovered agents (green) indicates that the epidemic continues to propagate over time. The total cumulative infections decrease to 447.9 ± 30.0 , representing a relative reduction of approximately 9% compared to the baseline. In addition, the infectious peak is approximately halved (89.6 ± 13.5 , vs. 177.2 ± 17.7) and occurs later in time (33.7 ± 6.9 days). Thus, lockdown measures primarily flatten and delay the epidemic curve rather than reduce the eventual attack rate, which is consistent with empirical observations during actual COVID-19 lockdowns [Hadjidemetriou et al. 2020; Shi et al. 2025; Vinceti et al. 2020].

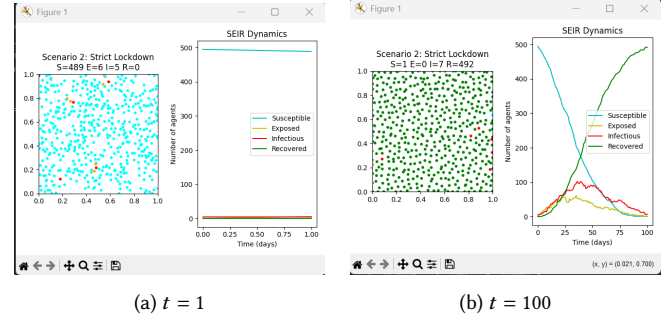


Fig. 2. Scenario 2 (Strict Lockdown, $\beta = 0.5$, $\varepsilon = 0.01$, $r = 0.05$, $\phi = 0$): spatial state at $t = 1$ (left) and $t = 100$ (right). Reduced mobility slows but does not prevent the outbreak.

5.3 Scenario 3: High Mask Compliance

A reduction in the per-contact transmission probability from $\beta = 0.5$ to $\beta = 0.1$ results in a similar reduction in overall disease burden (443.2 ± 18.0 infections, as shown in Figure 3), but with a substantially different temporal pattern: the peak of infectious cases is delayed significantly to $t \approx 48.5 \pm 6.2$ days, which is more than twice the baseline peak time. The panel at day 100 shows a residual cluster of recovered agents, while much of the periphery remains susceptible. Although mask usage and lockdown measures achieve similar final attack rates in this model, mask usage serves as a stronger temporal damper; the epidemic curve remains flatter for a longer duration. This mechanistic distinction supports the empirical finding that high levels of mask compliance significantly reduce the risk of SARS-CoV-2 infection in both the community and healthcare settings [Chu et al. 2020; Li et al. 2021; MacIntyre and Chughtai 2020].

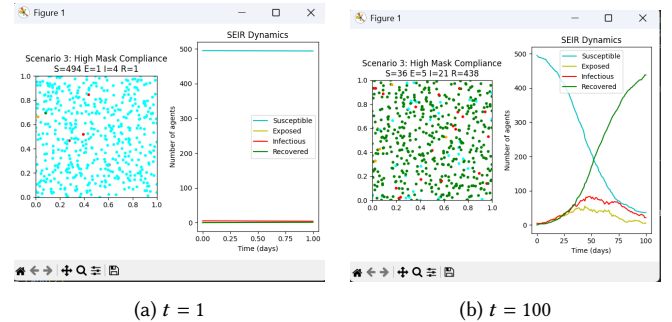


Fig. 3. Scenario 3 (High Mask Compliance, $\beta = 0.1$, $\varepsilon = 0.03$, $r = 0.05$, $\phi = 0$): spatial state at $t = 1$ (left) and $t = 100$ (right). Reduced per-contact transmission probability flattens and delays the epidemic curve.

5.4 Scenario 4: Social Distancing

Reducing the effective contact radius from $r = 0.05$ to $r = 0.02$ narrows the spatial window within which transmission may occur (Figure 4). At $t = 100$ the susceptible fraction is intermediate between Scenarios 1 and 3, but the total cumulative infections of 365.9 ± 67.3 indicate that distancing is the most effective single-parameter intervention in terms of total burden, with the peak

shifted to $t \approx 61.1 \pm 6.9$ days, representing the latest peak among any of the single-mechanism scenarios. The substantially larger standard deviation (± 67.3) reflects increased sensitivity to the spatial arrangement of seeded infectious agents under a reduced contact radius: when initial infectious seeds are located near population clusters, local outbreaks still develop; otherwise, transmission remains sparse. The radius-reduction approach is methodologically consistent with network-based formulations of distancing [Maheshwari and Albert 2020] and aligns with empirical and theoretical findings on the role of interpersonal distance in respiratory droplet transmission [de Souza Melo et al. 2021; Prakash et al. 2022; Qian and Jiang 2020; Sun and Zhai 2020; Thu et al. 2020].

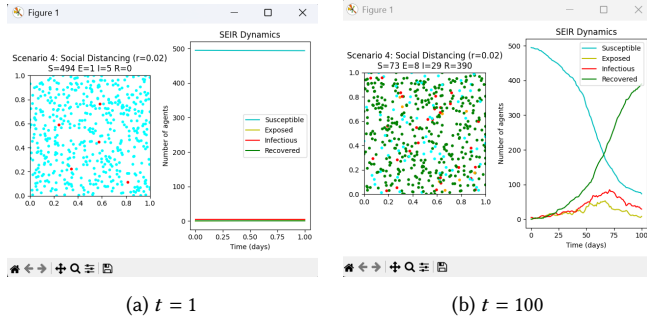


Fig. 4. Scenario 4 (Social Distancing, $\beta = 0.5$, $\varepsilon = 0.03$, $r = 0.02$, $\phi = 0$): spatial state at $t = 1$ (left) and $t = 100$ (right). A narrower effective contact radius produces the largest reduction in cumulative infections of any single-mechanism intervention.

5.5 Scenario 5: Vaccination Campaign

Initializing 40% of the population in the Recovered compartment alters both the initial and final configurations of the system (Figure 5). The day-1 panel displays a substantial green fraction alongside the cyan-colored susceptible majority and the five red seeds. By $t = 100$, a smaller fraction of the population has been newly infected. Cumulative new infections total 293.8 ± 1.4 (excluding the 200 vaccinated agents who began in the Recovered state). The infectious peak of 92.0 ± 12.0 occurs at $t = 25.9 \pm 3.0$ days. Although the peak height is comparable to those observed under lockdown and mask scenarios, the absolute number of new infections is the lowest among all single-intervention scenarios, demonstrating the indirect protection conferred by reducing the density of susceptible individuals in the population. The very small standard deviation indicates that pre-existing immunity functions as a deterministic damper on the epidemic. This finding is consistent with population-level immunity estimates from later phases of the COVID-19 pandemic [Moore et al. 2025; Pooley et al. 2023; Song et al. 2024].

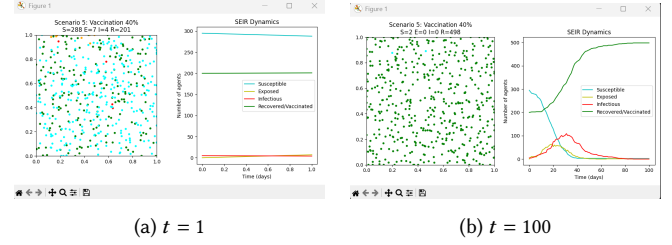


Fig. 5. Scenario 5 (Vaccination, $\beta = 0.5$, $\varepsilon = 0.03$, $r = 0.05$, $\phi = 0.40$): spatial state at $t = 1$ (left) and $t = 100$ (right). Forty percent of the population is initialised in the Recovered (green) state, lowering the susceptible density.

5.6 Scenario 6: Combined Intervention

The combined implementation of all four interventions ($\beta = 0.1$, $\varepsilon = 0.01$, $r = 0.02$, $\phi = 0.40$) results in effective containment of the outbreak (Figure 6). At $t = 100$, the population state is nearly indistinguishable from that at $t = 1$: the susceptible region remains dominated by cyan, the 40% pre-existing recovered fraction is maintained, and the original five infectious seeds have transitioned to the Recovered state without sustained onward transmission. Cumulative new infections decrease to 3.3 ± 3.2 , an order of magnitude lower than any single-intervention scenario, and the peak number of infectious agents is only 5.3 ± 0.6 , essentially the peak number alone. This result demonstrates the established multiplicative effect of layered non-pharmaceutical and pharmaceutical interventions and serves as the literature-informed reference point for the optimisation in Section 5.8.

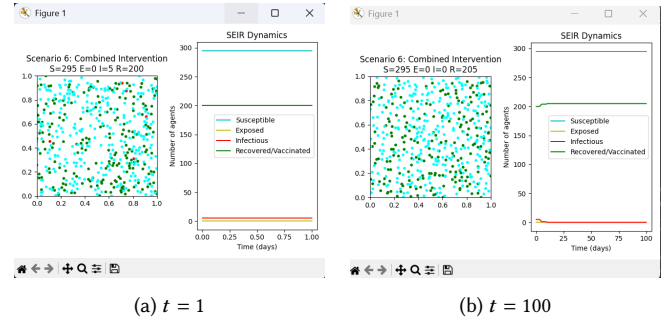


Fig. 6. Scenario 6 (Combined Intervention, $\beta = 0.1$, $\varepsilon = 0.01$, $r = 0.02$, $\phi = 0.40$): spatial state at $t = 1$ (left) and $t = 100$ (right). The end state is nearly indistinguishable from the initial configuration; the outbreak is essentially contained.

5.7 Cross-Scenario Comparison

Tables 2 and 3 summarise the parameter settings and aggregate outcome metrics for the six fixed-parameter scenarios, respectively. The numbered first column in both tables enables cross-referencing each scenario configuration with its observed dynamics. Two notable patterns emerge. First, while lockdown and mask interventions achieve comparable final attack rates, masks delay the epidemic peak by approximately fifteen days compared to lockdown: identical end-states reached through distinct temporal trajectories. Second, social

distancing alone (Scenario 4) yields the largest reduction in cumulative infections among the single-mechanism scenarios, but with the highest variance, indicating that its outcome depends strongly on the spatial arrangement of seeded cases. Vaccination (Scenario 5) achieves the lowest single-mechanism new-infection count with very low variance, consistent with the deterministic dampening effect of reduced susceptible density. Combined interventions (Scenario 6) reduce the total burden by approximately two orders of magnitude relative to baseline, motivating a search for whether a near-optimal parameter combination can do even better.

Table 2. Parameter sets for fixed-parameter Scenarios 1–6.

#	Scenario	β	ϵ	r	Init R
1	Baseline	0.50	0.030	0.05	0%
2	Lockdown	0.50	0.010	0.05	0%
3	Masks	0.10	0.030	0.05	0%
4	Distancing	0.50	0.030	0.02	0%
5	Vaccination	0.50	0.030	0.05	40%
6	Combined	0.10	0.010	0.02	40%

Table 3. Aggregate metrics for fixed-parameter Scenarios 1–6, averaged over 30 independent simulation seeds. Values reported as mean $\pm$ standard deviation. Total infections excludes the 40% Recovered population initialised in Scenarios 5 and 6.

#	Scenario	Total infections	Peak I	Time to peak
1	Baseline	494.9 $\pm$ 0.3	177.2 $\pm$ 17.7	21.5 $\pm$ 2.4
2	Lockdown	447.9 $\pm$ 30.0	89.6 $\pm$ 13.5	33.7 $\pm$ 6.9
3	Masks	443.2 $\pm$ 18.0	92.2 $\pm$ 12.8	48.5 $\pm$ 6.2
4	Distancing	365.9 $\pm$ 67.3	70.8 $\pm$ 14.9	61.1 $\pm$ 6.9
5	Vaccination	293.8 $\pm$ 1.4	92.0 $\pm$ 12.0	25.9 $\pm$ 3.0
6	Combined	3.3 $\pm$ 3.2	5.3 $\pm$ 0.6	2.1 $\pm$ 4.8

5.8 Scenario 7: Genetic Algorithm Optimisation

The genetic algorithm was applied in 30 independent runs. Each run consisted of 8 individuals over 15 generations, with a unique random seed assigned to each run. Figure 7 presents the convergence behavior. It also provides a comparison of SEIR dynamics between the parameters from the literature-informed Scenario 6 and the overall best solution identified by the genetic algorithm.

The convergence panel in Figure 7 (left) demonstrates that all 30 runs converge to a stable region of the search space within approximately 5 generations, with the mean trajectory reduced to fewer than 5 cumulative infections by generation ≈ 7 . The shaded band (± 1 standard deviation across runs) narrows rapidly as the number of generations increases, indicating that although the initial populations differ across runs, the genetic algorithm reliably identifies the same low-infection region of parameter space.

The comparison panel in Figure 7 (right) overlays the Infectious and Exposed trajectories from Scenario 6 and the genetic algorithm-identified optimum on the same axes. Both interventions contain the outbreak; however, the genetic algorithm optimum maintains the

number of infectious individuals at approximately zero throughout the simulation period, whereas Scenario 6 produces a small residual infectious tail.

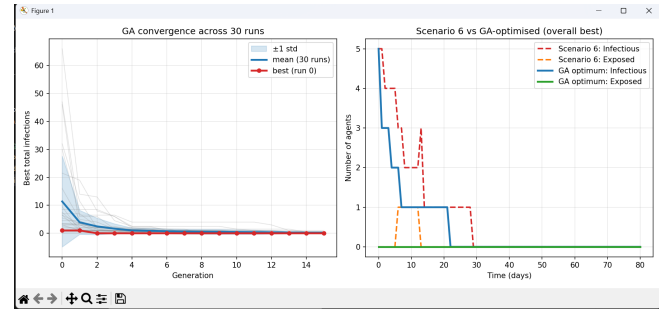


Fig. 7. Scenario 7 (Genetic Algorithm Optimisation). **Left:** GA convergence across 30 independent runs. Faint grey traces show individual runs; the blue curve and shaded band show the mean and ± 1 standard deviation across runs; the red trace highlights the overall best run. **Right:** comparison of Infectious (red) and Exposed (orange/green) trajectories under the literature-informed Scenario 6 parameters (dashed) and the GA-identified optimum (solid).

Table 4 presents a quantitative comparison of the literature-informed Scenario 6 parameters against the GA mean $\pm$ standard deviation across 30 runs. The GA converges to $\beta = 0.0723 \pm 0.0180$, $\epsilon = 0.0153 \pm 0.0077$, $r = 0.0236 \pm 0.0031$, and $\phi = 0.4312 \pm 0.1682$, with a mean total infection count of 0.2 ± 0.3 , which is essentially zero and an order of magnitude lower than the Scenario 6 mean of 3.3 ± 3.2 . The best solution from each run across all 30 runs is reported in Table 5 for completeness. All runs result in at most a single infection beyond the initial seed cases.

Table 4. Aggregate results of 30 independent GA runs (8 individuals $\times$ 15 generations each), compared against the literature-informed Scenario 6 parameters. Values reported as mean $\pm$ standard deviation.

Parameter	Description	Scenario 6	GA Mean $\pm$ Std
β	Transmission prob.	0.10	0.0723 $\pm$ 0.0180
ϵ	Mobility	0.010	0.0153 $\pm$ 0.0077
r	Transmission radius	0.02	0.0236 $\pm$ 0.0031
ϕ	Initial immune frac.	0.40	0.4312 $\pm$ 0.1682
Infections	Mean total infections	3.3 $\pm$ 3.2	0.2 $\pm$ 0.3

A notable characteristic of the GA-identified optimum is the asymmetric convergence of parameters. The two contact-related parameters (β and r) converge closely, with relative standard deviations of 25% and 13% of their respective means. This result indicates that the GA strongly favors low transmission probability and a small contact radius. In contrast, mobility (ϵ) and initial immunity (ϕ) display much broader variability, with relative standard deviations of 50% and 39%, respectively. These results imply that various combinations of mobility and initial immunity produce similar outcomes, provided that β and r are sufficiently constrained. Therefore, the optimum appears to reside along a ridge in the (ϵ, ϕ) -subspace, where mobility reduction and vaccination coverage function as partially

Table 5. Best solution found in each of the 30 independent GA runs. The overall best solution (run 0) is highlighted in bold.

Run	β	ε	r	ϕ	Infections
0	0.0591	0.0237	0.0206	0.0099	0.0
1	0.0797	0.0197	0.0200	0.5432	0.0
2	0.0546	0.0155	0.0260	0.5823	0.0
3	0.0732	0.0076	0.0222	0.5874	0.0
4	0.0614	0.0055	0.0328	0.4475	0.0
5	0.0770	0.0270	0.0223	0.4664	0.5
6	0.0984	0.0117	0.0233	0.3833	0.0
7	0.0519	0.0200	0.0211	0.5917	0.5
8	0.0796	0.0288	0.0275	0.5642	0.0
9	0.0615	0.0180	0.0261	0.5271	0.5
10	0.0669	0.0116	0.0213	0.5720	0.0
11	0.0769	0.0079	0.0217	0.0947	0.0
12	0.0541	0.0186	0.0201	0.3620	0.5
13	0.0644	0.0292	0.0237	0.4587	0.5
14	0.1086	0.0261	0.0229	0.3809	1.0
15	0.0856	0.0094	0.0290	0.5808	0.5
16	0.0816	0.0105	0.0222	0.1841	0.0
17	0.1182	0.0074	0.0225	0.5531	0.5
18	0.0867	0.0233	0.0214	0.5497	0.0
19	0.0579	0.0051	0.0219	0.4650	0.5
20	0.0578	0.0088	0.0236	0.3572	0.0
21	0.0634	0.0203	0.0221	0.5619	0.0
22	0.0613	0.0116	0.0203	0.4823	0.0
23	0.0502	0.0268	0.0273	0.5516	0.0
24	0.0544	0.0076	0.0266	0.4556	0.5
25	0.0739	0.0222	0.0210	0.1555	0.0
26	0.1102	0.0063	0.0235	0.5540	0.5
27	0.0818	0.0118	0.0254	0.5709	0.0
28	0.0521	0.0094	0.0204	0.0871	0.0
29	0.0660	0.0069	0.0288	0.2562	0.5

substitutable mechanisms once per-contact transmission risk and effective contact radius have been minimized.

6 Discussion

The experimental campaign indicates that all four single-mechanism interventions modeled in this study reduce the cumulative attack rate compared to the baseline, though they exhibit distinct temporal patterns and total burdens. Lockdown and mask interventions result in similar final attack rates (approximately 89% of the population in both cases), but differ significantly in the timing and magnitude of the infectious peak; masks delay the peak by approximately fifteen days relative to lockdown, despite achieving the same final outcome. Social distancing, represented as a reduction in the effective transmission radius r , yields the largest reduction in cumulative infections among the non-pharmaceutical interventions, but also displays the highest variance, indicating strong sensitivity to the spatial distribution of initial cases. Vaccination, defined as a 40% pre-existing immune fraction, results in the lowest absolute number of new infections due to the indirect effect of reduced susceptible density and exhibits very low variance, consistent with the deterministic reduction associated with removing individuals from the transmission pool. The combined intervention decreases cumulative infections by approximately two orders of magnitude relative to baseline, while the GA-optimized solution further reduces infections to a mean of 0.2 ± 0.3 , indicating that the literature-informed multi-layered intervention, although highly effective, does not represent the near-optimal point in the four-dimensional parameter space.

6.1 Asymmetric Parameter Convergence

An informative result from the genetic-algorithm analysis is the asymmetric convergence pattern observed across the four optimized parameters. The two contact-related parameters (β and r) converge closely across the thirty independent genetic algorithm runs, with relative standard deviations of 25% and 13%, respectively. In contrast, the mobility and immunity parameters (ε and ϕ) display substantially greater variability, with relative standard deviations of 50% and 39%. This pattern suggests that the loss landscape features a low-dimensional ridge in the (ε, ϕ) subspace, where mobility reduction and pre-existing immunity function as partially substitutable mechanisms, provided that per-contact transmission risk and effective contact radius are minimized. From a policy perspective, this finding implies that resource trade-offs between vaccination campaigns and mobility restrictions may be more flexible than mechanistic intuition alone would suggest, whereas benefits from biological transmission control (masks) and behavioral distancing (proximity reduction) are not interchangeable with either approach.

6.2 Comparison with Prior Work

The observed results align with previous agent-based and network-based COVID-19 modeling studies. [Castro et al. 2022] and [Rodríguez et al. 2022] report similar curve-flattening effects resulting from social-distancing interventions in their multi-agent and reopening policy models. Additionally, [Wang et al. 2024] demonstrates that hybrid cellular automata and agent-based approaches preserve the qualitative shape of intervention-response curves observed in this study. The integration of evolutionary optimization with epidemiological simulation extends the work of [Stanovov et al. 2022], who applied evolutionary algorithms to identify transmission mechanisms from first-wave data, and [Zheng et al. 2023], who optimized vaccine distribution under evolving demand profiles. In comparison to these studies, the present research provides a direct empirical comparison between the literature-informed combined intervention (Scenario 6) and the genetic algorithm (GA)-identified optimum (Scenario 7), quantifying the difference between empirically observed multi-layered policy and the mathematically optimal control point within this agent-based model (ABM). The per-gene robustness analysis across thirty GA runs further identifies which intervention mechanisms are considered essential by the optimization process and which are regarded as substitutable.

6.3 Limitations and Hidden Factors

The parameter values derived from the literature serve as a robust foundation for assessing the relative effectiveness of the four interventions. However, the model omits several real-world factors that affect both individual scenario predictions and the difference between Scenarios 6 and 7. The following sections identify key assumptions within the current model, summarize empirical evidence regarding discrepancies between these assumptions and observed realities, and propose extensions to address these gaps.

Vaccine hesitancy and effective coverage. First, the vaccination scenario assumes that 40% of the population is uniformly and effectively immune at the start of the simulation, thereby abstracting from the

reality that nominal vaccine availability does not directly translate into actual coverage. Empirical studies indicate that COVID-19 vaccine misinformation is both widespread and consequential. [Lee et al. 2022] report that 57.6% of surveyed US adults had encountered conspiratorial vaccine claims, with structural-equation analysis demonstrating that such exposure significantly increased hesitancy and reduced behavioral intention to vaccinate. [Pierri et al. 2022], using over 55 million tweets and CDC uptake data, find that geographic variation in online misinformation predicts roughly a 20% drop in COVID-19 vaccine uptake between US states, with Granger-causal evidence indicating that misinformation precedes hesitancy by a few days. [Zimmerman et al. 2023] document that self-reported vaccination intent in the United States fell from 74% to 56% within eight months, primarily due to non-medical misinformation concerning personal freedom, COVID denial, and unforeseen future effects, rather than medical concerns. The disparity between nominal and effective coverage is even more pronounced in lower-resource settings: [Ahmed et al. 2024] shows that first-dose coverage in Yemen reached only 2.6% by early 2022, with 61% of surveyed adults refusing the vaccine due to conspiracy theories and rumors. The long-term consequences are not hypothetical: [Orsini et al. 2022] uses Italian ICU admission data to demonstrate an order-of-magnitude difference in severe outcomes between vaccinated and unvaccinated populations, drawing a direct historical parallel with the 1960s polio campaign, during which hesitancy persistently reduced effective coverage below the politically achievable level. Future extensions of the present model could therefore replace the fixed scalar ϕ with a heterogeneous distribution stratified by demographic group and conditional on local exposure to misinformation channels, rather than treating uptake as a uniform population-wide parameter.

Behavioural compliance and trust in institutions. Beyond the immunity assumption, all four non-pharmaceutical intervention (NPI) scenarios in this study assume uniform compliance across the entire population: each agent moves with reduced mobility, exhibits a lowered per-contact transmission probability, and maintains a reduced contact radius. However, empirical research demonstrates that real-world compliance is partial and heterogeneous, and influenced by social factors beyond the standard SEIR specification. [Bargain and Aminjonov 2020] uses a cross-region panel of 233 European regions to show that non-essential mobility under identical lockdown stringency is reduced by approximately 5 percentage points more in high-trust regions than in low-trust regions. This effect intensifies with greater policy stringency and is reflected in lower growth in COVID-19 mortality. [Brodeur et al. 2021] synthesizes converging evidence that compliance with stay-at-home orders, mask wearing, and distancing measures is stratified by income, occupational exposure, broadband access, partisan media exposure, and interpersonal trust, so that nominally identical policies produce systematically unequal adherence across communities. [Petherick et al. 2021], in a global longitudinal study of 124 countries, document that adherence to costly distancing behaviours declined by approximately 0.68 points on a 5-point scale within 150 to 180 days of the first measure, even when policy stringency remained constant. This decline was mitigated in higher-trust societies. Consequently, the simulated

effectiveness of each intervention in the present study represents an upper bound. The difference between the literature-informed combined intervention (Scenario 6) and the genetic algorithm (GA)-identified optimum (Scenario 7) may be substantially reduced in real-world contexts where compliance is incomplete and unevenly distributed. Future research could replace the population-wide parameters β , ϵ , r , and ϕ with agent-specific compliance distributions conditional on socio-economic and trust-related covariates.

Hidden social structure and spatial heterogeneity. Beyond individual compliance, the simulation domain itself is an abstraction: a uniform unit square in which agents follow random walks and infection occurs whenever a susceptible agent enters the transmission radius of an infectious agent. While this abstraction effectively captures average-case contact behavior, it ignores the strong clustering that empirical contact data consistently reveal. [Mosson et al. 2008], in the population-based POLYMOD diary survey of 7,290 participants across eight European countries, document an average of 13.4 contacts per person per day, with strong age- and location-based assortativity. Contacts are concentrated in households, schools, workplaces, and leisure settings, and daily contact counts vary by an order of magnitude between age groups. [Prem et al. 2017] extend these contact matrices to 152 countries and demonstrate that location-specific mixing patterns (home, school, work, and other) significantly alter age-structured epidemic outcomes and the predicted impact of interventions such as school closure. Network-based simulations make the same point mechanistically. [Eubank et al. 2004], using a population-scale contact graph for the city of Portland derived from census and mobility data, find that the underlying location network is scale-free with a small number of high-degree hubs, while the person-contact graph exhibits small-world properties with clustering several orders of magnitude above an Erdős-Rényi baseline. [Kim et al. 2023], fitting both an Erlang-SEIR ODE and a scale-free agent-based model to South Korean contact-tracing data (19,123 infectors and 44,621 infectees), shows that the empirical out-degree distribution is fat-tailed, with 10% of infectors causing 41% of cluster transmissions, a pattern not reproduced by the homogeneous-mixing model. [Luo et al. 2022] reports from a hierarchical Wuhan community-network model that household contacts alone account for approximately half of all SARS-CoV-2 transmissions, and [Nagarajan et al. 2020] quantifies from Karnataka contact-tracing records that under 1% of infected individuals were responsible for approximately 38% of secondary cases. [Di Lauro et al. 2021] formalize the underlying theoretical point, demonstrating that homogeneous-mixing models systematically misestimate herd-immunity thresholds and intervention impacts compared with models that explicitly incorporate degree heterogeneity, household clustering, and age-structured mixing matrices. A future extension of the present model could therefore replace the uniform random-walk domain with a structured contact graph calibrated to country-specific home, school, work, and other contact matrices. This approach would allow the spatial radius r to serve as one component of a multi-layer network, rather than the sole representation of human proximity.

Heterogeneous agents and super-spreader dynamics. In the present model, all agents are assigned identical transmission and recovery

parameters. However, empirical evidence indicates that COVID-19 transmission is highly over-dispersed: a small proportion of cases accounts for the majority of secondary infections, and such cluster dynamics are not adequately represented by homogeneous- β models. [Sun et al. 2021], through contact-tracing analysis of 1,178 SARS-CoV-2 cases and 15,648 close contacts in Hunan, China, estimated a dispersion parameter of $k = 0.30$ (95% CI 0.23 to 0.39) and demonstrated that 80% of secondary infections originated from only 15% of primary infectors. [Wang et al. 2021] synthesised 60 dispersion estimates across SARS, MERS, and COVID-19, finding that 90% of estimated k values are below 1 and arguing that compartmental or agent-based models (ABMs) employing a single β systematically understate the potential for resurgence at low effective reproduction numbers. [Wegehaupt et al. 2023], in a systematic meta-analysis of 28 COVID-19 dispersion studies, reported that 93% of studies yielded $k < 1$ and emphasised that superspreading regimes result in outbreaks that are both more explosive and more responsive to targeted interventions, rather than population-wide measures, a dynamic that homogeneous- β models cannot capture. The current uniform- β formulation therefore likely underestimates both the peak infectious burden, due to the absence of super-spreader-driven surges, and the stochasticity in the tail of the epidemic, as outbreaks depend disproportionately on a small subset of highly connected agents. Future model extensions could assign each agent a β value drawn from a heavy-tailed distribution calibrated to the empirical dispersion values described above, enabling the genetic algorithm (GA) to optimise not only the mean per-contact risk but also the appropriate level of inter-agent variance.

Compliance fatigue and time-varying interventions. Finally, in the temporal dimension, the current model maintains the intervention parameters β , ε , r , and ϕ as constants throughout the 80-day simulation period. However, both empirical and theoretical studies indicate that compliance declines significantly over time, thereby substantially influencing epidemic trajectories. [Rypdal et al. 2020] extends the classical SIR framework by incorporating an exponential-decay term in the social-response rate ($v_3 \approx 0.1$ to 0.3 per day for European countries), demonstrating that models with constant parameters cannot reproduce a secondary wave that exceeds the first, regardless of the timing of relaxation. [Delussu et al. 2022], employing mixed-effects regression on Facebook and Google mobility data in relation to Italy's tiered restriction system from November 2020 to May 2021, find that mobility-based adherence declined at a baseline rate of approximately 0.08% per day and at twice that rate (0.16% per day) under the strictest red tier. This decline led to an approximate 15% rebound in mobility over seven months and revealed a clear stringency-adherence trade-off, with stricter policies experiencing more rapid declines. [N'konzi et al. 2022] formalize this trade-off mathematically by integrating an SEIR compartmental model with an imitation-dynamics ODE for adherence, showing that models assuming constant compliance systematically overestimate the benefits of stringent NPIs, and that incorporating endogenous adherence produces multiple epidemic waves rather than the single-peak trajectory observed in static-parameter simulations. Consequently, the constant-parameter assumption in the present study likely overstates the sustained benefits of interventions beyond the initial

weeks, and the disparity between the literature-informed combined intervention and the GA-identified optimum may increase further when compliance fatigue is considered. Future research could address this limitation by replacing the static gene set (β , ε , r , ϕ) with time-dependent functions calibrated to mobility-trace data, potentially modeling decay as an exponential function of intervention duration, consistent with [Rypdal et al. 2020] and [Delussu et al. 2022].

When considered together, these five abstractions tend to shift the simulated output toward correlation-related rather than independent ways. Vaccine hesitancy and partial compliance both reduce the *effective* application of every intervention. Super-spreader heterogeneity and clustered social structure introduce variance and concentrated transmission events that uniform-mixing models cannot generate. Compliance fatigue erodes the interventions the model assumes are sustained. Therefore, the simulated effectiveness of every scenario in this study should be interpreted as an upper bound under idealized conditions. The apparent advantage of the GA-optimized solution over the literature-informed combined point may diminish substantially once realistic combinations of these frictions are incorporated.

6.4 Methodological Caveats

Beyond the empirical limitations previously discussed, several methodological choices further constrain the generalisability of these findings. The simulation population is fixed at $N = 500$ agents, which is appropriate for a proof-of-concept but remains several orders of magnitude smaller than empirical agent-based COVID-19 studies that simulate between 10^4 and 10^6 individuals [Aleta et al. 2020; Eubank et al. 2004]. Consequently, finite-size fluctuations are more pronounced than in realistic populations, which partly explains the high-variance regime observed under social distancing (Scenario 4). The spatial domain is modeled as a uniform unit square traversed by random walks, thereby abstracting the clustered contact topology described under Hidden Social Structure. This abstraction represents the principal structural limitation of the model. The simulation horizon is set at 80 days, which is sufficient to capture the peak and decay of an unmitigated outbreak, but is shorter than the multi-year timescales characteristic of actual pandemics. Extending the simulation horizon would further challenge the constant-parameter assumption due to increased compliance fatigue. Furthermore, the genetic-algorithm fitness surface flattens rapidly once the basic reproduction number falls below unity. The GA-identified optimum of 0.2 ± 0.3 infections is located in a sub-critical region where the epidemic cannot propagate, resulting in the optimisation identifying a stable plateau rather than a distinct optimum. Therefore, the reported GA values reflect the breadth of the sub-critical attractor as much as the precision of the optimisation procedure.

6.5 Implications for Policy

Three policy-relevant observations arise from the experiments. First, the absolute difference between the literature-informed combined intervention (Scenario 6) and the genetic algorithm (GA)-identified optimum (Scenario 7) is minimal in practical terms (3.3 vs. 0.2 cumulative infections). Achieving the GA-identified optimum requires

parameters that are unlikely to be attainable through realistic policy measures, indicating that the literature-informed scenario already captures most of the feasible benefit. Second, the asymmetric GA convergence demonstrates that strict per-contact risk reduction (masks) and strict contact-radius reduction (distancing) are not easily substitutable, whereas mobility restrictions and vaccination coverage are more interchangeable. This finding supports the empirical heuristic that masks and distancing should be implemented together rather than as alternatives. Third, the high variance observed under social distancing alone highlights a known characteristic of low-density transmission: when the contact radius is small, outcomes are highly sensitive to the spatial distribution of initial cases. This suggests that early case detection and targeted isolation become especially valuable once distancing measures are in place.

7 Conclusion

This study developed an agent-based SEIR simulation of COVID-19 transmission within the PyCX framework and evaluated six fixed-parameter intervention scenarios, alongside a seventh scenario that applied genetic-algorithm optimization across 30 independent runs. All four single-mechanism interventions reduced cumulative infections relative to the baseline; the literature-informed combined intervention reduced total infections by approximately two orders of magnitude, and the GA-identified optimum reduced them by an additional order of magnitude. The GA additionally exhibited asymmetric convergence, in which the contact-related parameters β and r converged tightly across runs, while the mobility and immunity parameters ε and ϕ spanned a wide range of near-equivalent solutions, suggesting partial substitutability between lockdown intensity and vaccination coverage along an optimal-policy ridge. These findings illustrate the value of combining mechanistic ABM simulation with evolutionary optimization for layered intervention analysis.

The code for this project is available at: GitHub (master branch).

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